

AS ON 17 JULY 2026

PART 2

EDITORIAL VOCABULARY ENRICHMENT
FOR UPSC CSE

THEME:

**CYBER SECURITY,
CRITICAL INFRASTRUCTURE
PROTECTION &
NUCLEAR SECURITY**



WHAT & WHY IN THE NEWS

- A reported ransomware-linked data leak involving the **Kudankulam Nuclear Power Plant** has renewed concerns regarding cybersecurity, critical infrastructure protection, and national security.
- The incident highlights the urgent need for robust **cyber hygiene**, strong **cyber resilience**, and protection of strategic installations from evolving cyber threats.

CYBER THREATS ARE REAL. PREVENT TODAY. PROTECT TOMORROW.



WHY IT MATTERS



1

CYBER HYGIENE

- English Meaning:** Best practices for maintaining digital security.
- Hindi Meaning:** डिजिटल सुरक्षा
- Marathi Meaning:** डिजिटल सुरक्षा
- Use:** Cyber hygiene promotes security practices.
- UPSC Explanation:** Protection of critical infrastructure is a core component of national security.

UPSC RELEVANCE:

GS II : International Relations
GS III : Internal Security



2

CYBER RESILIENCE

- English Meaning:** Ability to prevent, withstand and recover from cyberattacks.
- Hindi Meaning:** डिजिटल प्रतिरोध
- Marathi Meaning:** डिजिटल प्रतिरोध
- Use:** Cyber resilience strengthens digital security.
- UPSC Explanation:** Cyber resilience ensures continuity of critical services during disruptions.

UPSC RELEVANCE:

GS II : International Relations
GS III : Internal Security



3

RANSOMWARE

- English Meaning:** Malware that encrypts data and demands payment.
- Hindi Meaning:** फंदाई वॉरम (कंप्यूटर) / डाटा चोरी
- Marathi Meaning:** फंदाई वॉरम (कंप्यूटर) / डाटा चोरी
- Use:** Ransomware attacks threaten critical services.

UPSC RELEVANCE:

GS II : International Relations
GS III : Internal Security



VALUE ADDITION

IMPORTANT CONCEPTS

- Five Trusts: Confidentiality, Integrity, Availability
- Data Trust Architecture
- Incident Response Plan
- Threat Intelligence Sharing



KEY STAKEHOLDERS

- National Cyber Security Coordinator (NCSC)
- Nuclear Power Corporation of India Limited (NPCIL)
- CERT-In (Indian Computer Emergency Response Team)
- National Critical Information Infrastructure Protection Centre (NCIIPC)

LEGAL & POLICY FRAMEWORK



Information Technology Act, 2008 (amended)
"Legal backbone for cybercrime and cyberterrorism"



National Cyber Security Policy, 2013
"Steers security, resilience and trusted cyber-space"



Nuclear Safety & Security Guidelines
"NSSG guidelines for nuclear installations"

STAY UP-TO-DATE

10177 Free eBook (PDF) : <https://www.insightsias.com/free-ebook> | 2024-25: 27001 - 82188 (Shubham)



UPSC MAINS PRACTICE QUESTION (10 MARKS, 200 WORDS)

- Q. Cyberattacks on critical infrastructure pose serious challenges to national security. Discuss the importance of cyber resilience and cyber hygiene in protecting strategic installations such as nuclear power plants from ransomware attacks.

